



Interior Design Project Manager Database

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Project Description

- **The interior design project manager database stores details about interior design projects, furniture items, fabric materials, contractors, and the schedules of tasks assigned during the project lifecycle. It tracks how furniture items are linked to fabric swatches, how contractors are assigned to tasks, and how each project is organized and executed efficiently.**
- **Consider the following requirements list:**

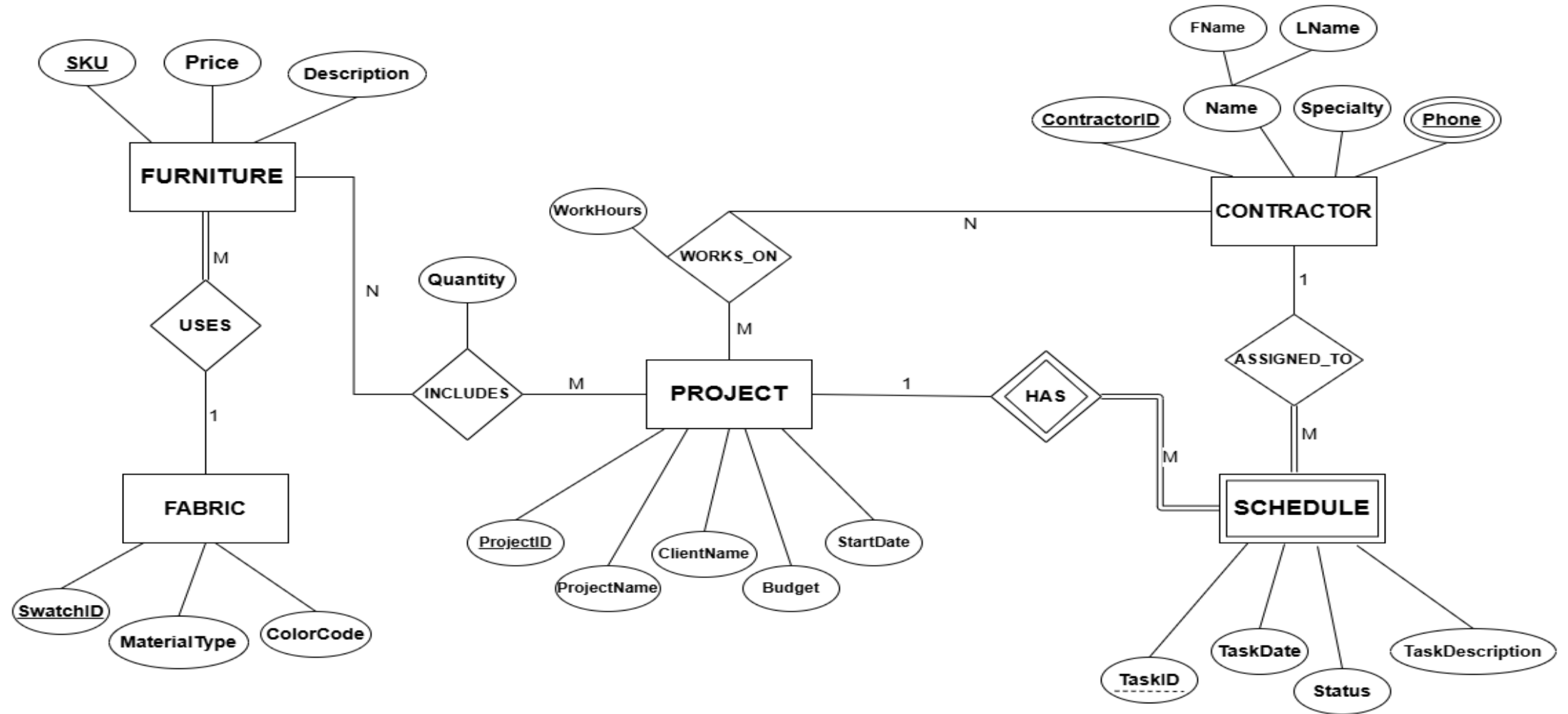
- The system manages one or more interior design projects.
- A project may include multiple furniture items, and a furniture item may be included in multiple projects.
- A furniture item must use exactly one fabric swatch,

- A fabric swatch can be used for multiple furniture items.
- Each project has a project identifier, project name, client name, budget, and start date.
- Each furniture item has a SKU, description and price.

Project Description

- The system must track the quantity of each furniture item included in a specific project.
- Each fabric swatch has a swatch identifier, material type, and color code.
- Contractors are responsible for executing tasks within projects. Each contractor has a contractor identifier, a composite name (first name and last name), phone numbers, and specialty.
- The system tracks the total work hours for each contractor on a project.
- Each project maybe has one or more scheduled tasks. Each task includes a description, task id, date, and status.
- A schedule is dependent on a project and cannot exist without it.
- Each task must be assigned to one contractor, while a contractor can handle multiple tasks.
- Furniture items must always be associated with a fabric swatch, and tasks must always be linked to both a project and a contractor.

Entity Relationship Diagram (ERD)



Relational Schema

PROJECT

<u>ProjectID</u>	ProjectName	ClientName	Budget	StartDate
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SCHEDULE

<u>TaskID</u>	<u>ProjectID</u>	TaskDate	Status	TaskDescription	ContractorID
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FURNITURE

<u>SKU</u>	Price	Description	SwatchID
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PROJECT_FURNITURE

<u>ProjectID</u>	<u>SKU</u>	Quantity
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FABRIC

<u>SwatchID</u>	MaterialType	ColorCode
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PROJECT_CONTRACTOR

<u>ProjectID</u>	<u>ContractorID</u>	WorkHours
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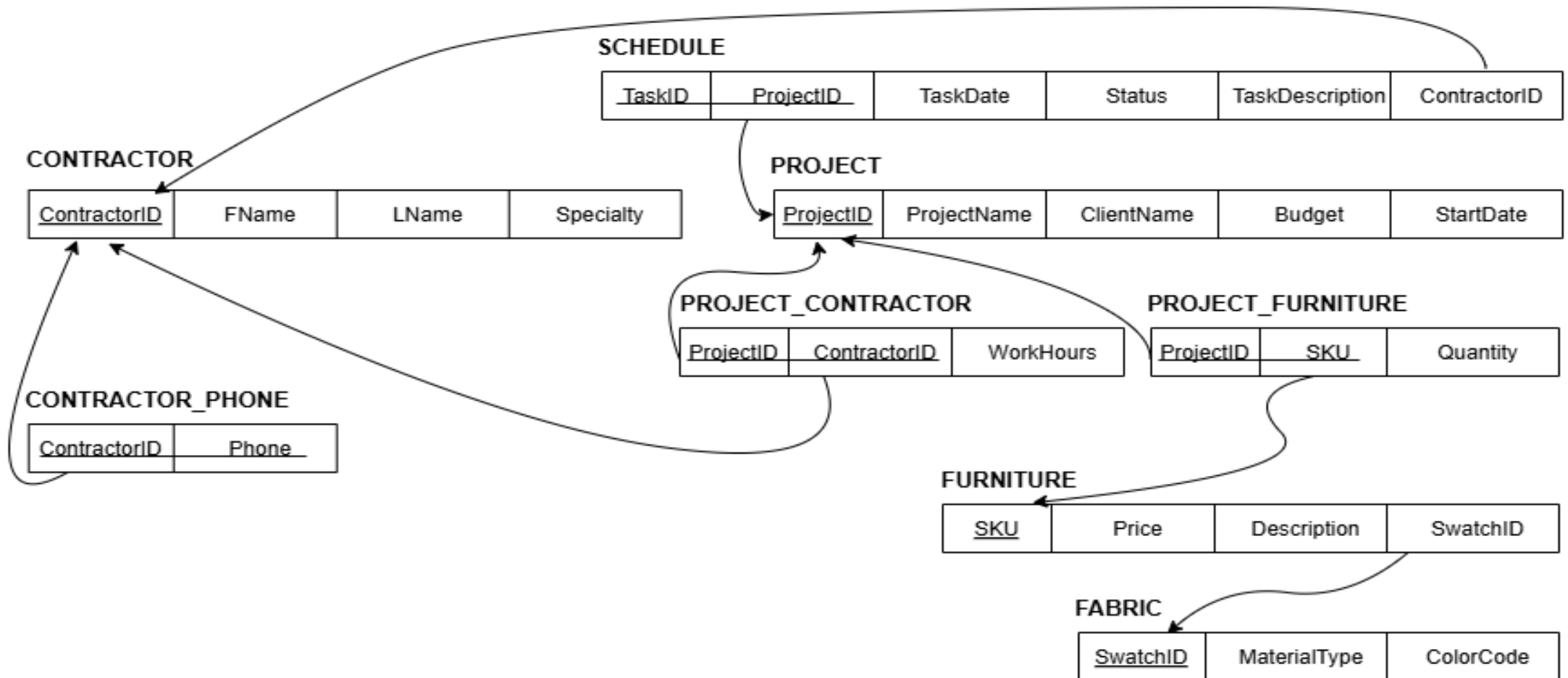
CONTRACTOR

<u>ContractorID</u>	FName	LName	Specialty
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CONTRACTOR_PHONE

<u>ContractorID</u>	<u>Phone</u>
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Relational Schema



DDL (Data Definition Language)

```
CREATE TABLE PROJECT (  
    ProjectID NUMBER(5) CONSTRAINT proj_id_pk PRIMARY KEY,  
    ProjectName VARCHAR2(50) NOT NULL,  
    ClientName VARCHAR2(20),  
    Budget NUMBER(10, 2),  
    StartDate DATE DEFAULT SYSDATE  
);  
CREATE TABLE FABRIC (  
    SwatchID NUMBER(5) CONSTRAINT Swatch_id_pk PRIMARY KEY,  
    MaterialType VARCHAR2(50) NOT NULL,  
    ColorCode VARCHAR2(50)  
);
```

DDL (Data Definition Language)

```
CREATE TABLE CONTRACTOR (  
    ContractorID NUMBER(5) CONSTRAINT cont_id_pk PRIMARY KEY,  
    FName VARCHAR2(20) NOT NULL,  
    LName VARCHAR2(20) NOT NULL,  
    Specialty VARCHAR2(50)  
);  
CREATE TABLE FURNITURE (  
    SKU VARCHAR2(20) CONSTRAINT sku_pk PRIMARY KEY,  
    Description VARCHAR2(200),  
    Price NUMBER(8, 2),  
    SwatchID NUMBER(5),  
    CONSTRAINT Swatch_id_fk FOREIGN KEY (SwatchID)  
        REFERENCES FABRIC (SwatchID)  
);
```

DDL (Data Definition Language)

```
CREATE TABLE CONTRACTOR_PHONE (  
    ContractorID NUMBER(5),  
    Phone VARCHAR2(20),  
    CONSTRAINT phone_pk PRIMARY KEY (ContractorID, Phone),  
    CONSTRAINT phone_cont_fk FOREIGN KEY (ContractorID)  
        REFERENCES CONTRACTOR (ContractorID)  
);  
  
CREATE TABLE SCHEDULE (  
    TaskID NUMBER(5),  
    ProjectID NUMBER(5),  
    TaskDescription VARCHAR2(200),  
    TaskDate DATE,  
    Status VARCHAR2(20),  
    ContractorID NUMBER(5),  
    CONSTRAINT sched_pk PRIMARY KEY (TaskID, ProjectID),  
    CONSTRAINT sched_proj_fk FOREIGN KEY (ProjectID)  
        REFERENCES PROJECT (ProjectID),  
    CONSTRAINT sched_cont_fk FOREIGN KEY (ContractorID)  
        REFERENCES CONTRACTOR (ContractorID)  
);
```

DDL (Data Definition Language)

```
CREATE TABLE PROJECT_FURNITURE (  
    ProjectID NUMBER(5),  
    SKU VARCHAR2(20),  
    Quantity NUMBER(3) CONSTRAINT Quantity_check CHECK (Quantity > 0),  
    CONSTRAINT pf_pk PRIMARY KEY (ProjectID, SKU),  
    CONSTRAINT pf_proj_fk FOREIGN KEY (ProjectID)  
        REFERENCES PROJECT (ProjectID),  
    CONSTRAINT pf_sku_fk FOREIGN KEY (SKU)  
        REFERENCES FURNITURE (SKU)  
);  
  
CREATE TABLE PROJECT_CONTRACTOR (  
    ProjectID NUMBER(5),  
    ContractorID NUMBER(5),  
    WorkHours NUMBER(5, 2),  
    CONSTRAINT pc_pk PRIMARY KEY (ProjectID, ContractorID),  
    CONSTRAINT pc_proj_fk FOREIGN KEY (ProjectID)  
        REFERENCES PROJECT (ProjectID),  
    CONSTRAINT pc_cont_fk FOREIGN KEY (ContractorID)  
        REFERENCES CONTRACTOR (ContractorID)  
);
```